

1. Title of the invention

Method and System for Energy-Optimal, Risk-Bounded Autonomous Actuation in Ambient-Intelligence IoT Using a Conformal Sensing Cascade with Optimised Risk-Budget Allocation (Aml-CC)

2. Field / Area of invention

Ambient Intelligence (Aml) for AI-native Internet of Things; context-aware multimodal sensing; energy-efficient edge AI; trustworthy (statistically guaranteed) autonomous actuation in smart homes, buildings, assisted living and Industry 5.0. (Responds to the IEEE IoT Journal special-issue CFP on "Ambient Intelligence for AI-Native IoT: Invisible, Context-Aware, and Autonomous Experiences": topics – Aml architectures, context-aware sensing, edge AI, explainable/trustworthy Aml, energy-efficient AI-native IoT.)

3. Prior patents and publications from literature

#	Reference (verify before filing)	What it teaches	Gap vs. present invention
1	US 9,159,208, "Energy efficient cascade of sensors for automatic presence detection" (USPTO)	Cascade of sensors, cheaper sensors gate costlier ones to save power.	Gate is heuristic/threshold based; no distribution-free bound on wrong-actuation; no budget split; no deferral semantics.
2	Angelopoulos, Bates, Fisch, Lei, Schuster, "Conformal Risk Control", ICLR 2024 (arXiv:2208.02814)	Calibrating a single threshold to bound expected monotone loss.	Single model/stage; not a multi-sensor, energy-costed cascade; no allocation of a global budget across stages.
3	"Fast yet Safe: Early-Exiting with Risk Control", arXiv:2405.20915	Risk-controlled early exit in neural networks.	Exits inside one network (compute saving); does not power physical sensor tiers, no energy-optimal per-tier budget split, no actuate/escalate/defer policy.
4	"Efficient Conformal Prediction via Cascaded Inference with Expanded Admission", arXiv:2007.03114	Cascaded models to cheapen conformal sets.	Model cascade for set size; no sensing-hardware wake-up, no actuation risk.
5	Gibbs & Candès, "Adaptive Conformal Inference Under Distribution Shift", NeurIPS 2021	Online adaptation of miscoverage level under shift.	Generic; not applied to per-tier budgets of a sensing cascade.
6	Jitkrittum et al., "When Does Confidence-Based Cascade Deferral Suffice?", NeurIPS 2023 (arXiv:2307.02764)	Analysis of confidence-threshold deferral in model cascades.	Heuristic confidence rules, no finite-sample guarantee (our baseline behaves this way).
7	"Sensor-Aware Classifiers for Energy-Efficient Time Series Applications on IoT Devices", arXiv:2407.08715	Early-exit classifiers exploiting sensor structure for energy.	No statistical guarantee on wrong decisions; no joint budget allocation.
8	"Early Time Classification with Accumulated Accuracy Gap Control", arXiv:2402.00857	Stopping-time classification with accuracy-gap control.	Time-series stopping, not multi-tier hardware sensing, no energy-optimal allocation.

Note: a professional patent-database search (Espacenet/Google Patents/InPASS, CPC G06N 20/00, G16Y, H04W 52/02, G05B 13) must confirm this table before filing. References 1–8 were identified by web search on 30-Sep-2026; only titles/abstract-level content was reviewed.

4. Summary and background of the invention (gap / novelty)

Background. Ambient IoT must act *invisibly* (no user prompts), yet a wrong autonomous action (unlocking a door, switching a heater, dismissing a fall alert) is costly, while always-on rich sensors (camera, mmWave radar) drain batteries and raise privacy exposure. Existing systems either (a) run all sensors and a single classifier (high energy, and softmax confidence is uncalibrated), or (b) use heuristic confidence-threshold sensor cascades that give no assurance on how often they act wrongly, especially for unseen occupants.

Gap. No known system couples a *physical sensing-tier wake-up cascade* with a *finite-sample, distribution-free bound on wrong autonomous actuation* while *choosing how that bound is spent across tiers to minimise energy*.

Invention. At each tier a split-conformal prediction set over candidate intents is computed. A singleton set triggers actuation and stops (higher-energy sensors stay asleep); a non-singleton set wakes the next tier; ambiguity at the last tier defers to the user (or a safe default). A wrong actuation implies the true intent was outside the stopping tier's set, so the wrong-actuation probability is bounded by the sum of per-tier miscoverage levels (union bound) – the **risk budget** α . The novel **budget-allocation step** searches the split of α over tiers that minimises expected energy + λ -deferral-rate on one data split, and calibrates the thresholds on a *disjoint* split, so the guarantee stays valid. On-site recalibration with few labelled samples restores the bound under occupant shift.

5. Objective(s) of the invention

(i) Bound the rate of erroneous autonomous actuation by a user/operator-set budget α without distributional assumptions; (ii) minimise sensing energy by waking high-power / privacy-intrusive modalities only when cheaper ones are ambiguous; (iii) provide a principled abstain ("ask user") behaviour instead of forced decisions; (iv) make the guarantee portable to new occupants/homes through lightweight recalibration; (v) run on edge hardware (logistic/linear heads suffice; cost of allocation is a small grid search).

6. Working principle (in brief)

Tier- t score $s=1-p_t(y|x)$; threshold $q_t = \text{ceil}((n+1)(1-\alpha_t))$ -th smallest calibration score; set $C_t(x)=\{k: 1-p_{t,k}(x) \leq q_t\}$. Rule: $|C_t|=1 \rightarrow$ actuate & stop; else escalate; last tier non-singleton $\rightarrow$ defer. $P(\text{wrong actuation}) \leq \sum \alpha_t \leq \alpha$. Allocation: $(\alpha_{1..T})^* = \text{argmin}$

$E[\text{energy}] + \lambda P(\text{defer})$ s.t. $\sum \alpha_t \leq \alpha$ (grid search on allocation split), thresholds from a disjoint calibration split.

7. Description of the invention in detail

7.1 Architecture. (a) Sensing tiers T1 passive low-power (PIR, door contact, ambient light, time-of-day; 1 energy unit), T2 medium (plug power, audio level, BLE RSSI; 6 units), T3 high (camera / mmWave embedding; 40 units); (b) per-tier classifiers over cumulative features (edge MCU / gateway); (c) conformal calibrator storing per-tier score quantiles; (d) budget allocator; (e) actuation controller with escalate / actuate / defer outputs; (f) optional on-site recalibration module fed by user overrides.

7.2 Procedure. *Offline:* 1) train tier classifiers on training occupants; 2) on allocation split run the cascade for each candidate split $(\alpha_1, \dots, \alpha_T)$ with $\sum \alpha_t = \alpha$ and keep the split minimising $J = E[\text{cumulative energy}] + \lambda \cdot P(\text{defer})$; 3) on a disjoint calibration split compute q_t . *Online:* power tier 1, compute set; if singleton $\rightarrow$ actuate; else power tier 2 ... ; at the last tier defer if not singleton.

7.3 Why the bound holds. If the cascade actuates wrongly at tier t then $C_t = \{\mathbf{x}\}$ with $\mathbf{x} \neq y$, hence $y \notin C_t$; split-conformal gives $P(y \notin C_t) \leq \alpha_t$ under exchangeability; union over tiers gives $\leq \sum \alpha_t$. Because the allocation uses data disjoint from the calibration data, the q_t remain valid.

7.4 Implementation. Reference implementation in Python (repository: *amicg*: simulator.py, cascade.py; experiments/run_experiments.py; tests/test_cascade.py, unit tests pass). Drawing (block diagram):

Sensors T1 (1u)	$\rightarrow$	Classifier ₁ + conformal set C_1	$\rightarrow$	$ C_1 =1$? yes $\rightarrow$ ACTUATE (stop)
		no $\downarrow$ wake T2 (6u)		
Sensors T2 (+6u)	$\rightarrow$	Classifier ₂ + set C_2	$\rightarrow$	$ C_2 =1$? yes $\rightarrow$ ACTUATE
		no $\downarrow$ wake T3 (40u)		
Sensors T3 (+40u)	$\rightarrow$	Classifier ₃ + set C_3	$\rightarrow$	$ C_3 =1$? yes $\rightarrow$ ACTUATE ; no $\rightarrow$ DEFER to user

Fig. 1 – Conformal sensing cascade. Budget allocator chooses $\alpha_1 + \alpha_2 + \alpha_3 \leq \alpha$; wrong actuation $\leq \alpha$.

8. Experimental validation results

Setup (simulation; no real-deployment data yet). Synthetic multi-tier smart-home benchmark with 8 intents (idle, enter home, cooking, TV, sleep, leave, exercise, fall-risk), 3 cumulative sensing tiers (6/14/26 features, energy 1/6/40 units), per-occupant offsets (inter-occupant covariate shift) and deliberately confusable intent pairs at low tiers. Per seed: 24 training occupants, 10 allocation, 14 calibration, 24 unseen test occupants $\times$ 150 samples; 30 random seeds; mean values shown. Tier accuracies (single seed): T1 67%, T2 80%, T3 99.6%. Baselines: always-all-sensors argmax; tier-1 only; single-stage conformal with always-full sensing; confidence-threshold cascade (single threshold tuned on the same calibration split to hit α , forced decision at last tier); ablation: equal budget split.

α	Method	False-actuation rate	Energy / decision	Deferral rate	Seeds with FA> α
0.01	Aml-CC (proposed)	0.0067	32.3	0.009	7%
0.01	Aml-CC, equal split	0.0043	33.6	0.003	0%
0.01	Conformal, full sensing	0.0011	47.0	0.009	0%
0.01	Conf.-threshold cascade	0.0109	31.7	0.000	57%
0.01	All sensors, argmax	0.0032	47.0	0.000	0%
0.02	Aml-CC (proposed)	0.0141	29.8	0.009	7%
0.02	Aml-CC, equal split	0.0073	31.9	0.005	0%
0.02	Conformal, full sensing	0.0005	47.0	0.020	0%
0.02	Conf.-threshold cascade	0.0196	29.2	0.000	50%
0.02	All sensors, argmax	0.0032	47.0	0.000	0%
0.05	Aml-CC (proposed)	0.0417	24.1	0.002	10%
0.05	Aml-CC, equal split	0.0164	28.9	0.014	0%
0.05	Conformal, full sensing	0.0002	47.0	0.051	0%
0.05	Conf.-threshold cascade	0.0481	23.6	0.000	47%
0.05	All sensors, argmax	0.0032	47.0	0.000	0%
0.1	Aml-CC (proposed)	0.0900	17.6	0.002	17%
0.1	Aml-CC, equal split	0.0359	25.0	0.023	0%
0.1	Conformal, full sensing	0.0001	47.0	0.102	0%
0.1	Conf.-threshold cascade	0.0967	17.0	0.000	40%
0.1	All sensors, argmax	0.0032	47.0	0.000	0%

Table 1 – mean over 30 seeds (std. in results/results.json). Tier-1-only: FA=0.330, energy 1.

Findings. (1) The mean false-actuation rate of Aml-CC is below the budget for every α (e.g. 0.042 at $\alpha=0.05$; 0.090 at $\alpha=0.10$), as the bound predicts (per-seed violations, 7–17%, are finite-sample fluctuation of a bound that holds in expectation). (2) At $\alpha=0.05$

Aml-CC uses 24.1 energy units vs 47.0 for always-full sensing: **49% saving** (63% at $\alpha=0.10$: 17.6 vs 47.0) with FA 0.042 vs 0.0032 for the full-sensing argmax at its fixed operating point – i.e. the operator trades risk for energy in a controlled, tunable way. (3) Budget optimisation reduces energy by 17% vs equal split at $\alpha=0.05$ (24.1 vs 28.9) and allocates almost all the budget to tier 2 (mean split $\alpha_1/\alpha_2/\alpha_3 = [0.001, 0.046, 0.003]$). (4) The tuned confidence-threshold cascade reaches similar mean energy (23.6) but has no guarantee: it exceeds α in 47% of seeds at $\alpha=0.05$ (40–57% across α) versus 10% for Aml-CC, and cannot defer. **Honest limitation:** energy is on par with that heuristic baseline, the benefit is the guarantee and the defer option.

Robustness to occupant shift ($\alpha=0.05$). The exchangeability assumption fails when test occupants are more heterogeneous than calibration occupants: the bound is exceeded (Table 2). On-site recalibration with 800 labelled samples from the new environment restores it, paying with more deferrals/energy (the system safely asks more).

Occupant-shift severity	FA, calibrated pre-deployment	FA, + on-site recalibration (800 labelled)	Deferral after recalibration
$\times 1.0$	0.042	0.044	0.00
$\times 1.5$	0.062	0.041	0.02
$\times 2.0$	0.092	0.035	0.07
$\times 3.0$	0.178	0.027	0.32

Table 2.

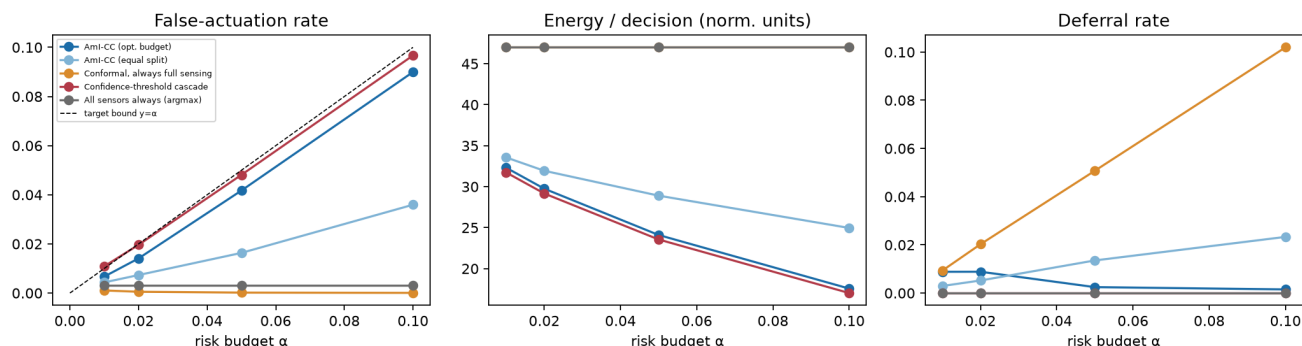


Fig. 2 – Risk-budget sweep: false-actuation, energy, deferral (30 seeds).

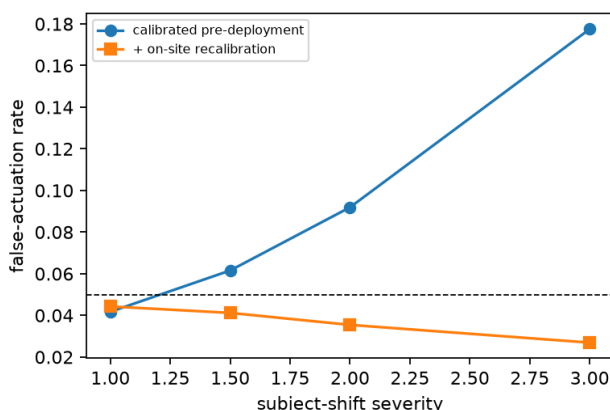


Fig. 3 – Effect of occupant shift with/without recalibration.

Reproduce: python experiments/run_experiments.py; python experiments/make_figures.py; python tests/test_cascade.py. **Status:** results are from a simulator, not field data; a pilot on a public smart-home dataset / campus testbed is planned before TRL₄ claims and is recommended before the IoTJ submission.

9. What aspect(s) need protection?

Independent claim 1 (method): controlling actuation in an ambient IoT system comprising: powering a first sensing tier; computing with a first classifier a conformal prediction set of candidate intents whose threshold is calibrated to a first miscoverage level α_1 ; actuating if the set is a singleton and otherwise powering a higher-energy tier and repeating; deferring if the last-tier set is not a singleton; wherein the miscoverage levels satisfy $\sum \alpha_i \leq \alpha$, a predetermined actuation-risk budget, and are selected by minimising an objective comprising expected sensing energy on a data split disjoint from the split used to calibrate the thresholds.

Dependent aspects: (2) objective includes deferral (user-interruption) cost λ ; (3) grid/convex search of the allocation; (4) per-tier classifiers on cumulative features; (5) on-site recalibration from user overrides when a drift detector fires, raising deferral to preserve the bound; (6) tiers including PIR/door/light $\rightarrow$ plug-power/audio/BLE $\rightarrow$ camera/mmWave, so privacy-intrusive sensors are powered only on ambiguity; (7) class-conditional (per-action criticality) budgets [not yet validated]; (8) system, and (9) non-transitory medium claims. *Technical effect emphasised for eligibility (e.g. Indian Patents Act s.3(k) / 35 USC §101): reduced power consumption of sensor hardware and a bounded rate of erroneous physical actuation.*

10. Technology readiness level

TRL 1	TRL 2	TRL 3	TRL 4	TRL 5	TRL 6	TRL 7	TRL 8	TRL 9
Basic principles observed	Concept formulated	[X] Experimental proof of concept (simulation, this work)	Validated in lab (planned: testbed)	Relevant environment	Demonstrated in relevant env.	Prototype in operational env.	System complete & qualified	Proven in operational env.

Ticked: **TRL 3** (experimental proof of concept). Target: TRL 4 after a lab testbed run.

----- END OF THE DOCUMENT -----